

Miguel Ángel Rodríguez García

OPTIMIZATION ENGINEER



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[GitHub](#)



[Personal web](#)

Education

MSc in Quantum Computing.

International University of La Rioja.

BSc in Computer Science.

University of Murcia.

BSc in Physics.

University of Murcia.

Completed 120 ECTS.

Languages

English: C1 Level

Projects and Courses

IBM Qiskit Global Summer School 2024

IBM Certificate in Qiskit framework for Quantum Computing.

Projects

Master's Thesis on A* quantum algorithm

- Researched the application of quantum computing principles to the shortest path problem, formulating a hybrid quantum-classical approach that leverages heuristic functions to guide quantum state propagation.
- Evaluated the theoretical complexity and feasibility of quantum search algorithms, demonstrating through modeling that the proposed approach can achieve near-100% probability in identifying optimal paths.

Parallelization Strategies for the A* algorithm

- Developed and implemented sequential and parallel versions of the A* heuristic search algorithm in C using OpenMP for shared-memory environments.
- Designed a decentralized search architecture based on Hash Distributed A*, achieving a significant reduction in execution time and up to a 2x speedup compared to the sequential version in complex graph scenarios.

HomeLab server

- Designed and deployed a custom home lab environment featuring a dedicated rack with structured cabling, managed switching, and hardware virtualization.
- Configured and maintained a high-availability server stack (Proxmox/Docker) to self-host critical services, implementing VLAN segmentation and robust network security protocols.

Experience

Embention Sistemas Inteligentes.

Nov 2025 - Now

Software Engineer responsible for reviewing and validating algorithm documentation for an Amazon project in the aerospace industry.

Private tutor in Physics, Mathematics and Chemistry

Sep 2017 - Dec 2025

Provided private tutoring in Mathematics, Physics, and Chemistry to students of all levels (primary, secondary, and high school), with a focus on academic support and exam preparation.

Master's Internship

Mar 2024 - May 2024

Donostia International Physics Center (DIPC)

Developed a Python script to solve the Bayesian Network Structure Learning (BNSL) problem using O'Gorman's algorithm.